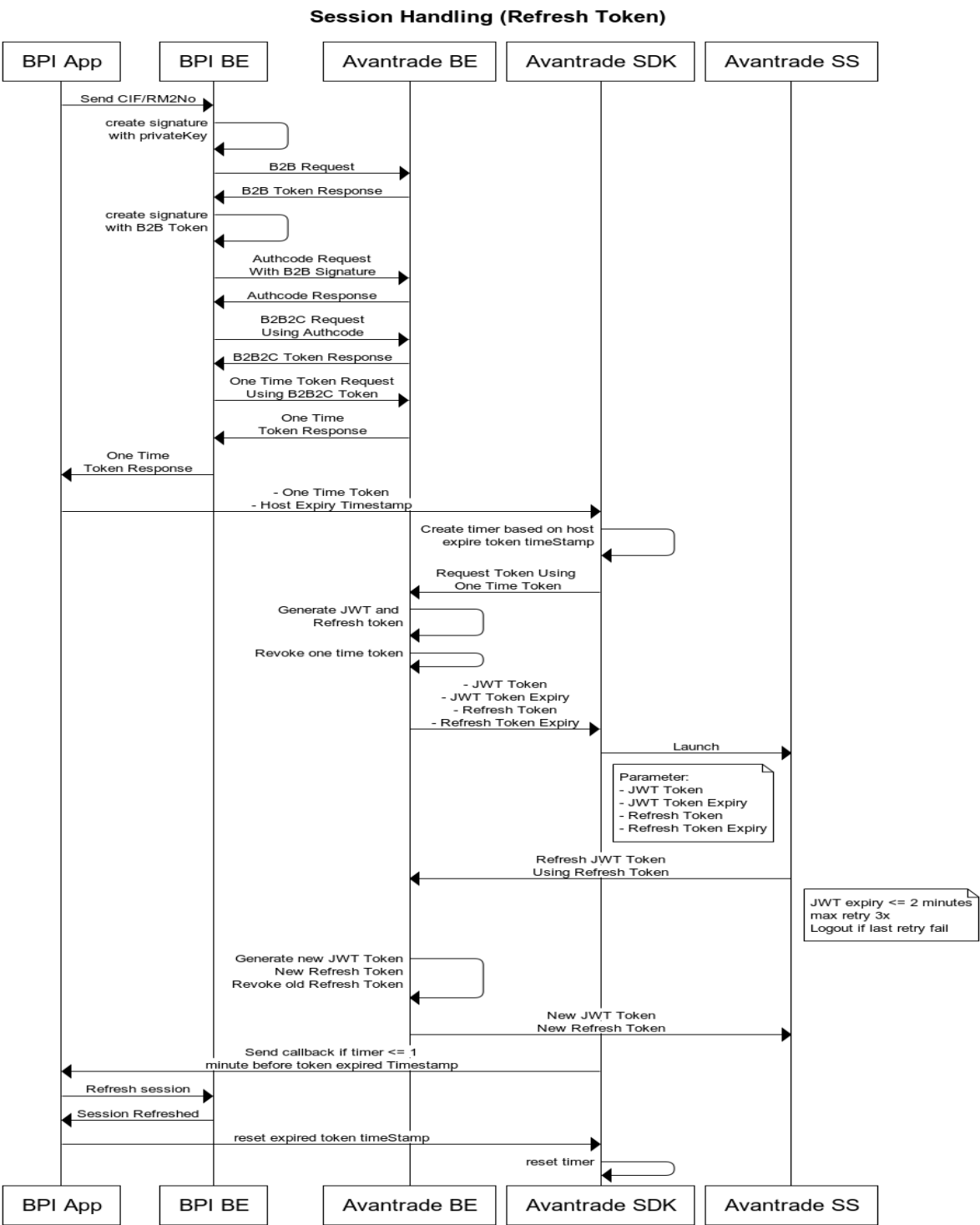


BPI Implementation

1. Session Handling Flow



2. B2B Request Signature

Ensures non-repudiation & integrity checking using asymmetric signature **SHA256withRSA**

(`Private_Key` , `stringToSign`).

- `Private_Key` is provided during the partner registration process

- `stringToSign = X-CLIENT-KEY + "|" + X-TIMESTAMP`

2.1 Function

```
public String generateAuthSignature(
    String clientId,
    String timestamp,
    String privateKeyPem) throws Exception {

    String cleaned = privateKeyPem
        .replace("-----BEGIN PRIVATE KEY-----", "")
        .replace("-----END PRIVATE KEY-----", "")
        .replace("-----BEGIN RSA PRIVATE KEY-----", "")
        .replace("-----END RSA PRIVATE KEY-----", "")
        .replaceAll("\\s", "");

    byte[] keyBytes = Base64.getDecoder().decode(cleaned);
    PrivateKey privateKey = KeyFactory.getInstance("RSA")
        .generatePrivate(new PKCS8EncodedKeySpec(keyBytes));

    String payload = clientId + "|" + timestamp;

    Signature signature = Signature.getInstance("SHA256withRSA");
    signature.initSign(privateKey);
    signature.update(payload.getBytes(StandardCharsets.UTF_8));

    return Base64.getEncoder().encodeToString(signature.sign());
}
```

2.1.1 Function Parameter

Key	Type	Length	Mandatory	Description	Sample Value
clientId	String	N.A	Y	Client UUID from Registration Process	53e75332-b619-4688-ad45-37d279110a16
timestamp	String	N.A	Y	TimeStamp when initiating process	2026-05-20T13:45:12.123+07:00
privateKeyPem	String	N.A	Y	Private Key from Registration Process	-----BEGIN PRIVATE KEY-----
					asdasgdjhg2jg21hj3g1jg3jh1g23hj12g
					-----END PRIVATE KEY-----

2.2 Usage

```
ZonedDateTime now = ZonedDateTime.now(TimeZone.getTimeZone("Asia/Jakarta").toZoneId());
String timeStamp = now.format(DateTimeFormatter.ofPattern("yyyy-MM-dd'T'HH:mm:ss.SSSXXX"));
String privateKeySignature;
try {
    privateKeySignature = helper.generateAuthSignature(
        constants.getClientId2(),
```

```
        timeStamp,  
        constants.getPrivateKeyPem2());  
} catch (Exception e) {  
    throw new CustomException(e.toString());  
}
```

3. B2B2C Request Signature

Represents the signature of a request using symmetric signature **HMAC_SHA512** (`secretKey` , `stringToSign`).

```
stringToSign = HTTPMethod + ":" + EndpointURL + ":" + access_token + ":" +  
Lowercase(HexEncode(SHA-256(requestBody))) + ":" + X-TIMESTAMP
```

If there is no request body, use an empty string `""`.

3.1 Function

```
public static String generateTransactionSignature(  
    String httpMethod,  
    String endPoint,  
    String token,  
    String timestamp,  
    String body,  
    String clientSecret) throws NoSuchAlgorithmException, InvalidKeyException {  
  
    String hexEncodeBody = sha256Hex(body);  
    String stringToSign = String.join(":", httpMethod, endPoint, token, hexEncodeBody, timestamp);  
  
    SecretKeySpec secretKey = new SecretKeySpec(clientSecret.getBytes(), "HmacSHA512");  
    Mac hmac = Mac.getInstance("HmacSHA512");  
    hmac.init(secretKey);  
  
    return Base64.getEncoder().encodeToString(  
        hmac.doFinal(stringToSign.getBytes(StandardCharsets.UTF_8)));  
}
```

3.1.1 Function Parameter

Key	Type	Length	Mandatory	Description	Sample Value
httpMethod	String	N.A	Y	endpoint method	POST
endPoint	String	N.A	Y	url endpoint	/auth-service/authentication/v2/access-token/b2b
token	String	N.A	Y	B2B Token	asdasgdjhg2jg21hj3g1jg3jh1g23hj12g
timestamp	String	N.A	Y	TimeStamp when initiating process	2026-05-20T13:45:12.123+07:00
body	String	N.A	Y	Empty String by default	
clientSecret	String	N.A	Y	channel id from B2B Token Response	53e75332-b619-4688-ad45-37d279110a16

3.2 Usage

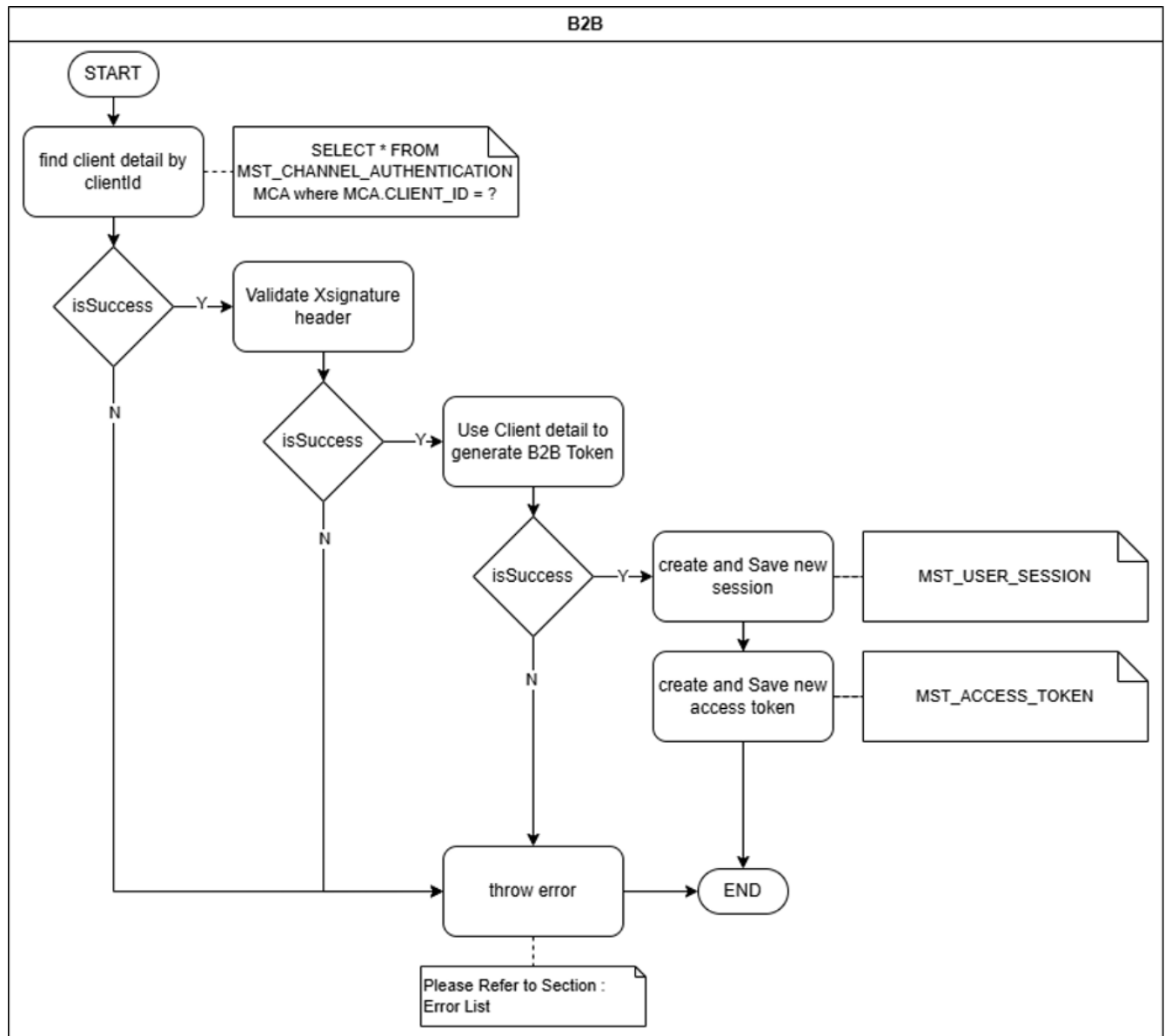
```
public String generateTransactionSignature(String timeStamp, UUID channelId, String b2bToken) {  
    try {
```

```
        return SignatureUtil.generateTransactionSignature(
            "POST",
            "/auth-service/authentication/v2/access-token/b2b",
            b2bToken,
            timeStamp,
            "",
            channelId.toString());
    } catch (Exception e) {
        throw new RuntimeException(e);
    }
}
```

4. API Detail

4.1 B2B

4.1.1 Flowchart



4.1.2 URL

POST api-gateway/auth-service/authentication/v2/access-token/b2b

4.1.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
X-CLIENT-KEY	String	N.A	Y	clientId from registration process	c04ee779-1a81-4453-83ea-fd370e9f4cc9
X-TIMESTAMP	String	N.A	Y	Timestamp when create secretKey and privateKey signature	7xDTUTKUIUF1e5XC9OsBxNJEN4C...
X-SIGNATURE	String	N.A	Y	privateKey Signature	SIGNATURE

4.1.4 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
grantType	String	N A	Y	-	client_credentials

4.1.5 Response

Property	Type Data	Sample Value
accessToken	String	8S1NjMEMF3lqQr2Q1PgPAN26j1aA
tokenType	String	Bearer
expiresIn	String	900 (Seconds)
sessionId	UUID	d568e0f0-740e-42fc-a65e-c8860a50cd3d
channelId	UUID	d568e0f0-740e-42fc-a65e-c8860a50cd3d
successCode	String	43
successMessage	String	Success

4.1.6 Data Relations

Property	Mapping	Relations
accessToken	OAuth2Token.getTokenValue	
tokenType	OAuth2AccessToken.getTokenType	
sessionId	MST_USER_SESSIONS.SESSION_ID	

4.1.7 Curl Example

```
curl --location '<host>/api-gateway/auth-service/authentication/v2/access-token/b2b' \
--header 'X-CLIENT-KEY: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'X-TIMESTAMP: 2025-07-11T14:21:23.456+08:00' \
--header 'X-SIGNATURE: cX5cBkPQwaXaV8qbRcwc/EmqI9spIEKrBQDCllsHgDD/AgqjS9Y9DQxkJX1UMHdy5z9L4YgrqSLvImD2HxwPfv2wAx0HBd+HENSx6TgX/YCzFex0jqUk12FML4B8wK13Eqp9ARbkrGAKhnlcUW+Q3IcTFM+miq+SUq0Q8j3vQjbBSKdpWgS2Qn0bPAu0BCXqmQaYkKZ7AP03bA/VKmkXi0ixq2H7/U8Y0qeX0K3hznYZnPZCMIkhJfamLXBHqDdLHUtiYoL3WaYTj+q+vv4Uah4Hj3I1zEUrkphR3lh10o3pF7EJUVqGYBrEYqJRKgBlT15T9z5UvEMbonGyCDK8Hw==' \
--header 'Content-Type: application/json' \
--data '{
```

```
"grantType": "client_credentials"
},'
```

4.1.1.8 Response Example

```
{
  "accessToken": "eyJhbGciOiJIUzI1NiJ9.dummy.payload.signature",
  "tokenType": "Bearer",
  "expiresIn": "900",
  "sessionId": "550e8400-e29b-41d4-a716-446655440000",
  "channelId": "WEB",
  "additionalInfo": {}
}
```

4.1.1.9 Error List

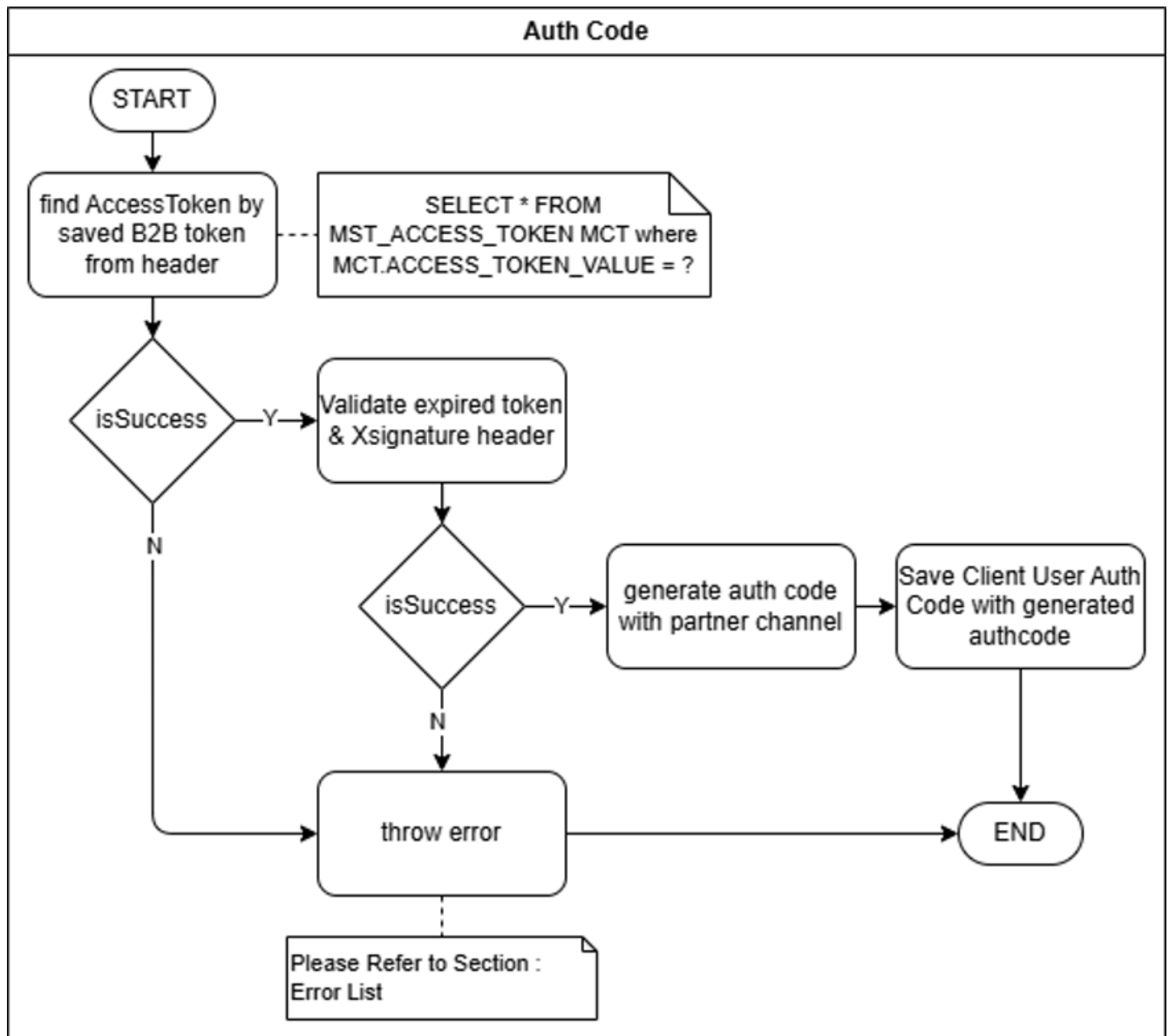
Messages	Message Code	HTTP Code
Token generation failed for client: \$clientKey	TOKEN_GENERATION_FAILED	401
Invalid Mandatory Field X-SIGNATURE	MISSING_SIGNATURE	401
Invalid Signature	INVALID_SIGNATURE	401
Invalid Mandatory Field X-TIMESTAMP	MISSING_TIMESTAMP	401
Invalid Format Field X-TIMESTAMP	INVALID_TIMESTAMP	401
Invalid X-CLIENT-KEY Not Match	INVALID_CLIENT_KEY	401
Invalid or missing X-Client-Key header.	MISSING_CLIENT_KEY	401
Generic Exception	GENERIC_EXCEPTION	500

4.1.1.10 Error Response Example

```
{
  "success": false,
  "errorCodes": ["TOKEN_GENERATION_FAILED"],
  "errors": ["Token generation failed for client: c04ee779-1a81-4453-83ea-fd370e9f4cc9"],
  "data": null,
  "messageCodes": null,
  "messages": null,
  "errorUid": "67de40be-08ef-4950-bbcd-83348c633c5f"
}
```


4.2 Auth Code

4.2.1 Flowchart



4.2.2 URL

POST api-gateway/auth-service/authentication/v2/generate-auth-code/b2b2c

4.2.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
X-PARTNER-ID	String	N.A	Y	clientId	c04ee779-1a81-4453-83ea-fd370e9f4cc9
X-TIMESTAMP	String	N.A	Y		7xDTUTKUIUF1e5XC9OsBxNJEN4C...

Key	Type	Length	Mandatory	Description	Sample Value
				Timestamp when create secretKey signature	
X-SIGNATURE	String	N.A	Y	secretKey Signature	SIGNATURE
X-BRANCH-ID	String	N.A	Y	clientId	c04ee779-1a81-4453-83ea-fd370e9f4cc9
Authorization	String	N.A	Y	B2B token	7xDTUTKUIUF1e5XC9OsBxNJEN4C...

4.2.4 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
paramType	String	N.A	Y	param type used	CIF_BANK
paramValue	String	N.A	Y	param value used for get customer data	CIF2501045138
partnerChannel	String	N.A	Y	registered partner channel	CH_BPI

4.2.5 Response

Property	Type Data	Sample Value
authCode	String	8S1NjMEMF3lqQr2Q1PgPAN26j1aA
successCode	String	43
successMessage	String	Success

4.2.6 Data Relations

Property	Mapping	Relations
authCode	MST_CLIENT_USER_AUTH_CODE.AUTH_CODE	

4.2.7 Curl Example

```
curl --location '<host>/api-gateway/auth-service/authentication/v2/generate-auth-code/b2b2c' \
--header 'Authorization: ato3DmdyH0nnnCKLB9jInGDmzMfPreV6BwUs8-6bYMBGg81Wx2ayEqHI1DRD7BRghPAhvs10eYnRW7bXL20_U4BCeR7yZSRb0blq5iTmuxOXXIZ-1u8rI1IX9uYUEZEH' \
--header 'X-PARTNER-ID: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'X-BRANCH-ID: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'X-TIMESTAMP: 2025-07-11T14:21:23.456+08:00' \
--header 'X-SIGNATURE: Ymk6zVb0Cj\hd9Jd7MdVYyD3DgZPnlpsFr8KgXB/d0z9Tmue6ul5UmpvBDqBKdKD0Jk9mKh0smBV4LiraJYvQ==' \
--header 'Content-Type: application/json' \
--data '{
  "paramType": "CIF_BANK",
  "paramValue": "000000001102859",
  "partnerChannel": "CH_BPI"
}'
```

4.2.8 Response Example

```
{
  "successCode": "43",
  "successMessage": "Success",
  "authCode": "AUTH-7F3A9C8D5B2E1F4A"
}
```

4.2.9 Error List

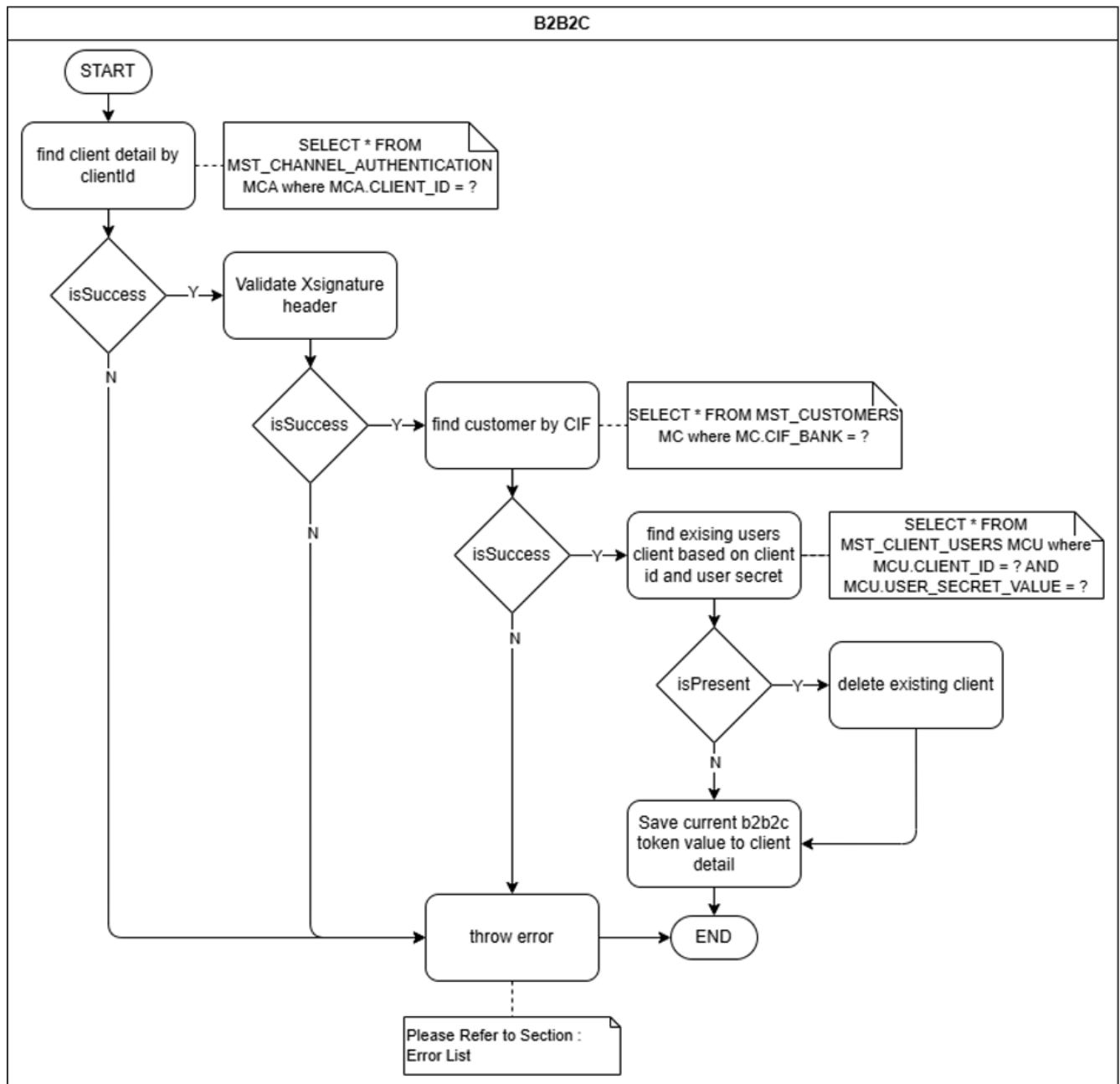
Messages	Message Code	HTTP Code
Token Not Found	TOKEN_INVALID	401
Token Expired	TOKEN_EXPIRED	401
Invalid Mandatory Field X-SIGNATURE	MISSING_SIGNATURE	401
Unauthorized. Invalid Client ID or Signature	INVALID_SIGNATURE	401
Invalid Mandatory Field X-TIMESTAMP	MISSING_TIMESTAMP	401
Invalid Format Field X-TIMESTAMP	INVALID_TIMESTAMP	401
Generic Exception	GENERIC_EXCEPTION	500

4.2.10 Error Response Example

```
{
  "success": false,
  "errorCodes": ["TOKEN_EXPIRED"],
  "errors": ["Token Expired"],
  "data": null,
  "messageCodes": null,
  "messages": null,
  "errorUId": "67de40be-08ef-4950-bbcd-83348c633c5f"
}
```

4.3 B2B2C

4.3.1 Flowchart



4.3.2 URL

POST api-gateway/auth-service/authentication/v2/access-token/b2b2c

4.3.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
X-CLIENT-KEY	String	N.A	Y	clientId	

Key	Type	Length	Mandatory	Description	Sample Value
					c04ee779-1a81-4453-83ea-fd370e9f4cc9
X-TIMESTAMP	String	N.A	Y	Timestamp when create secretKey and privateKey signature	7xDTUTKUIUF1e5XC9OsBxNJEN4C...
X-SIGNATURE	String	N.A	Y	privateKey Signature	SIGNATURE
X-BRANCH-ID	String	N.A	Y	clientId	c04ee779-1a81-4453-83ea-fd370e9f4cc9
AUTHORIZATION-CUSTOMER	String	N.A	Y	token from B2B Response	5207d4a3-5666-4d6d-8f1f-7732ca8fa85e

4.3.4 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
grantType	String	N.A	Y	grant type	AUTHORIZATION_CODE
authCode	String	N.A	Y	auth code from response	eyJhbGciOiJIUzUxMiJ9
refreshToken	String	N.A	Y	empty by default	-

4.3.5 Response

Property	Type Data	Sample Value
accessToken	String	B2B2C Token
tokenType	String	Bearer
accessTokenExpiryTime	String	1296000 (In Seconds - 15 Days)
refreshToken	String	8S1NjMEMF3lqQr2Q1PgPAN26j1aA
refreshTokenExpiryTime	String	900
responseCode	String	43
responseMessage	String	Success

4.3.6 Data Relations

Property	Mapping	Relations
accessToken	OAuth2Token.getTokenValue	
tokenType	OAuth2AccessToken.getTokenType	
accessTokenExpiryTime		
refreshToken		
refreshTokenExpiryTime		

4.3.7 Curl Example

```
curl --location '<host>/api-gateway/auth-service/authentication/v2/access-token/b2b2c' \
--header 'X-BRANCH-ID: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'X-CLIENT-KEY: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'X-TIMESTAMP: 2025-07-11T14:21:23.456+08:00' \
--header 'X-SIGNATURE: cX5cBkPQwaXaV8qbRcwc/EmqI9spIEKrBQDCllsHgDD/
AgqjS9Y9DQxkJX1UMHdy5z9L4YgrqSLvLmD2HxwPfV2wAx0HBd+HENsx6TgX/
YCzFex0jqUk12FML4B8wK13Eqp9ARbkrgAKhnlcUW+Q3IcTFM+miq+SUq0Q8j3vQjbBSKdpWgS2Qn0bPAu0BCXqmQaYkKZ7AP03bA/
VKmkXi0ixq2H7/
U8Y0qeX0K3hznYZnPZCMIkhJfamLXBHqDdLHUtiYoL3WaYTj+q+vv4Uah4Hj3I1zEUrkphR3lh10o3pF7EJUVqGYBrEyqJRKgBlT15T9z5
UvEMbonGyCDK8Hw==' \
--header 'AUTHORIZATION-CUSTOMER:
ato3DmdyH0nnnCKLB9jInGdmzMfPreV6BwUs8-6bYMBGg81Wx2ayEqHI1DRD7BRghPAhvs10eYnRW7bXL20_U4BCEr7yZSRb0blq5iTMux
0XKIZ-1u8rI1IX9uYUEZEH' \
--header 'X-PARTNER-ID: 47c5963a-4836-d939-e063-e30a1aac30ba' \
--header 'Content-Type: application/json' \
--data '{
  "grantType": "AUTHORIZATION_CODE",

  "authCode": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJDSUZfQkF0SyIsInVzZXIiOiIwMDAwMDAwMTEwMjg1OSIsImNoYW5uZWwiOiJDS
F9CUEkiLCJpYXQiOiJlE30DA50Dg5NTYsImV4cCI6MTc4MTM0MDk1Nn0.JEY7RqTyYvhZsRmtz_oIszRGdaliHic27v1glqy2uqGgyV9-
uQlagXQXTTP6QiQskOcPp1G83J8D6DqgtgV_wBg",
  "refreshToken": ""
}'
```

4.3.8 Response Example

```
{
  "responseCode": "2007300",
  "responseMessage": "Successful",
  "accessToken": "eyJhbGciOiJIUzI1NiJ9.dummy.access.token",
  "tokenType": "Bearer",
  "accessTokenExpiryTime": "1296000",
  "refreshToken": "",
  "refreshTokenExpiryTime": "",
  "additionalInfo": {}
}
```

4.3.9 Error List

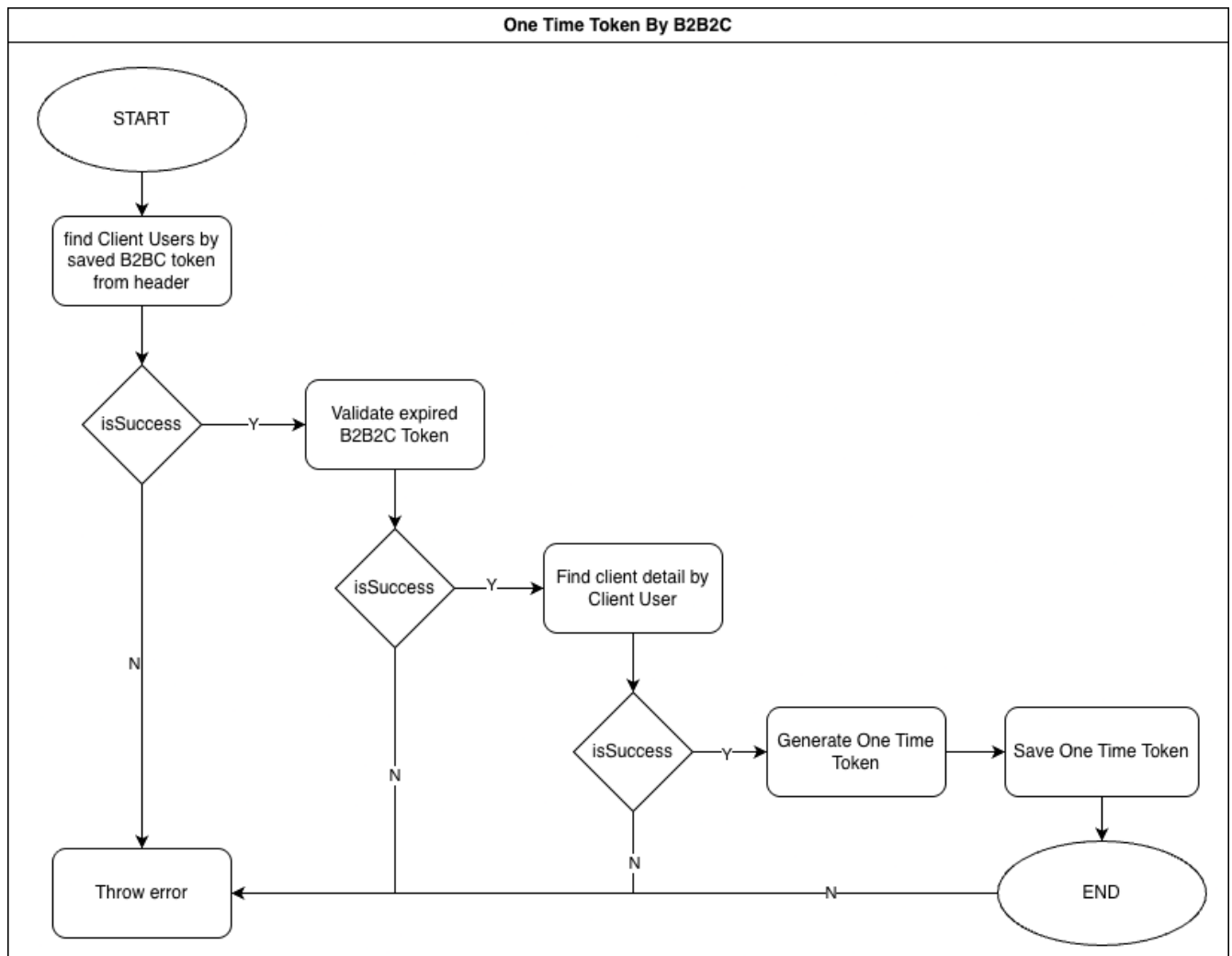
Messages	Message Code	HTTP Code
Invalid Authentication Code	INVALID_AUTH_CODE	401
Missing Authentication Code	MISSING_AUTH_CODE	401
Invalid Mandatory Field X-SIGNATURE	MISSING_SIGNATURE	401
Invalid Signature	INVALID_SIGNATURE	401
Invalid Mandatory Field X-TIMESTAMP	MISSING_TIMESTAMP	401
Invalid Format Field X-TIMESTAMP	INVALID_TIMESTAMP	401
User Not Found	USER_NOT_FOUND	401
Invalid X-CLIENT-KEY Not Match	INVALID_CLIENT_KEY	401
Invalid or missing X-Client-Key header.	MISSING_CLIENT_KEY	401
Generic Exception	GENERIC_EXCEPTION	500

4.3.10 Error Response Example

```
{
  "success": false,
  "errorCodes": ["INVALID_AUTH_CODE"],
  "errors": ["Invalid Authentication Code"],
  "data": null,
  "messageCodes": null,
  "messages": null,
  "errorUid": "67de40be-08ef-4950-bbcd-83348c633c5f"
}
```

4.4 One Time Token By B2B2C

4.4.1 Flowchart



4.4.2 URL

POST api-gateway/auth-service/authentication/v2/token/by-b2bc-token

4.4.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
authorization-customer	String	N.A	Y	B2B2C Token	7SZjVjOzWfm_VuPqilbieMAG8sh7UxhicaCz46h-GqmgTTIW5NXpxCS5dBCjuFw_AODJGoBg9XgFHY_IG5FkLzbYa
Authorization	String	N.A	Y	B2B2C Token	7SZjVjOzWfm_VuPqilbieMAG8sh7UxhicaCz46h-GqmgTTIW5NXpxCS5dBCjuFw_AODJGoBg9XgFHY_IG5FkLzbYa

4.4.4 Response

Property	Type Data	Sample Value
successCode	String	43
successMessage	String	Success
oneTimeToken	String	eyJhbGciOiJIUz...
oneTimeTokenExpiry	String	900

4.4.5 Data Relations

Property	Mapping	Relations
oneTimeToken	MST_ONE_TIME_TOKEN.TOKEN	
oneTimeTokenExpiry	MST_ONE_TIME_TOKEN.EXPIRED_AT	

4.4.6 Curl Example

```
curl -X POST <host>/api-gateway/auth-service/authentication/v2/token/by-b2bc-token \
-H "authorization-customer: <B2B2C_ACCESS_TOKEN>" \
-H "Authorization: <B2B2C_ACCESS_TOKEN>" \
-H "ENV: SIT"
```

4.4.7 Response Example

```
{
  "successCode": "43",
  "successMessage": "Success",
  "oneTimeToken": "eyJhbGciOiJIUzI1NiJ9..",
  "oneTimeTokenExpiry": "2026-06-04T09:35:49+07:00"
}
```

4.4.8 Error List

Messages	Message Code	HTTP Code
Missing B2BC Token	MISSING_TOKEN	401
Token Expired	EXPIRED_TOKEN	401
User Not Found	USER_NOT_FOUND	401
Generic Exception	GENERIC_EXCEPTION	500

4.4.9 Error Response Example

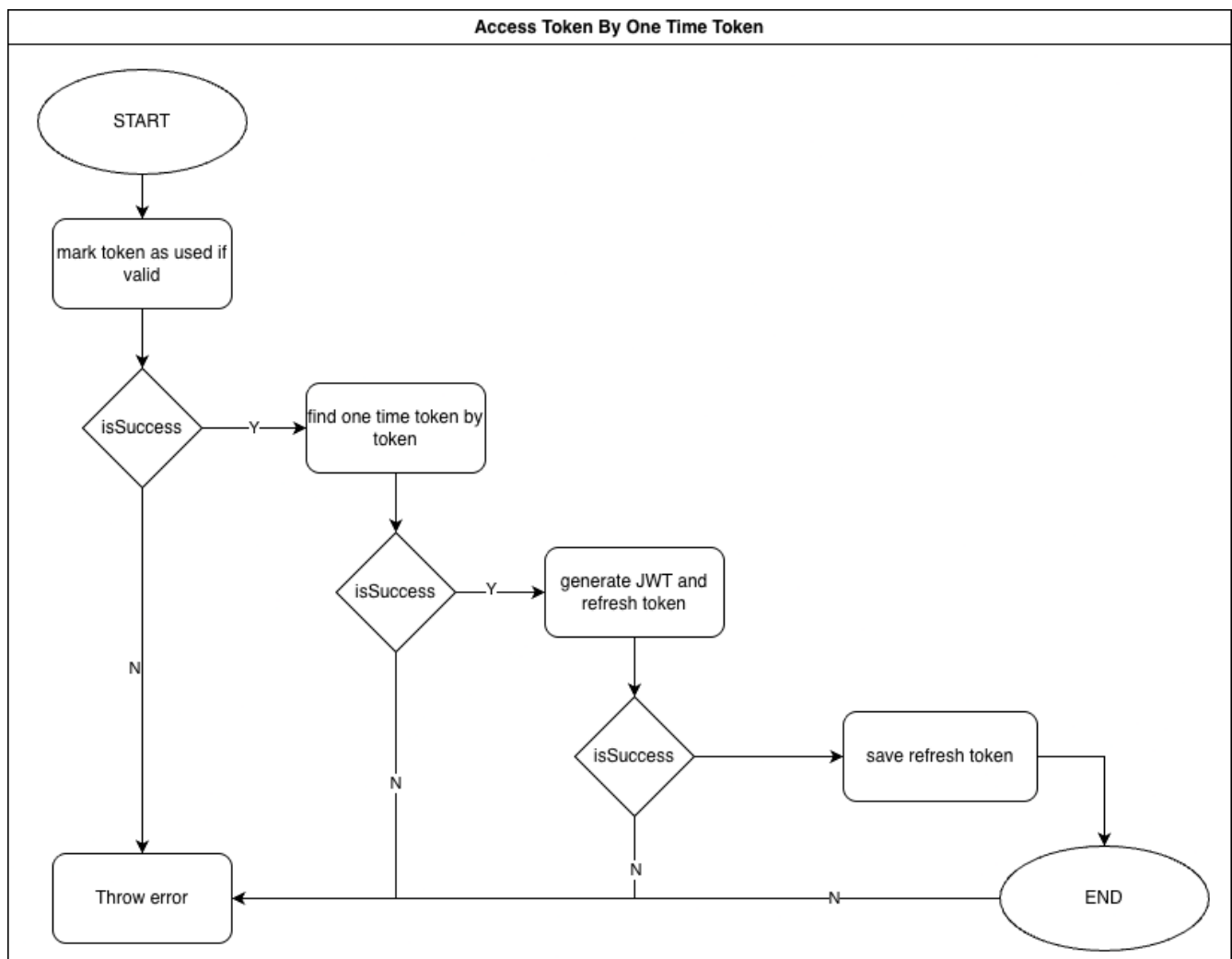
```
{
  "success": false,
  "errorCodes": ["MISSING_TOKEN"],
  "errors": ["Missing B2BC Token"],
  "data": null,
  "messageCodes": null,
  "messages": null,
}
```

```
"errorUid": "67de40be-08ef-4950-bbcd-83348c633c5f"
}
```

4.5 Access Token By One Time Token

Note: This API should only be hit by the **Avantrade side**. Host side should not generate JWT token — JWT token currently can only be generated by the **SDK**.

4.5.1 Flowchart



4.5.2 URL

POST api-gateway/auth-service/authentication/v2/token/by-one-time-token

4.5.3 Header Parameter

Key	Type	Length	Mandatory	Description	Sample Value
ENV	String	N.A	Y	Selected Environment (DEV/GAD/SIT/UAT)	DEV

4.5.4 Body Parameter

Key	Type	Length	Mandatory	Description	Sample Value
oneTimeToken	String	N.A	Y	One Time Token	eyJhbGciOiJIUzUxMiJ9...

4.5.5 Response

Property	Type Data	Sample Value
successCode	String	43
successMessage	String	Success
jwtToken	String	eyJhbGciOiJIUzUxMiJ9...
jwtTokenExpiry	String	900
refreshToken	String	eyJhbGciOiJIUzUxMiJ9...
refreshTokenExpiry	String	TimeStamp
refreshTokenExpiry	String	TimeStamp
webviewUrl	String	dynamic Webview URL

4.5.6 Data Relations

Property	Mapping	Relations
refreshTokenExpiry	MST_REFRESH_TOKEN.EXPIRED_AT	

4.5.7 Error List

Messages	Message Code	HTTP Code
Missing B2BC Token	MISSING_TOKEN	401
Token Expired	EXPIRED_TOKEN	401
User Not Found	USER_NOT_FOUND	401
Generic Exception	GENERIC_EXCEPTION	500

4.5.8 Error Response Example

```
{
  "success": false,
  "errorCodes": ["EXPIRED_TOKEN"],
  "errors": ["Token Expired"],
  "data": null,
  "messageCodes": null,
  "messages": null,
}
```

```
"errorUid": "67de40be-08ef-4950-bbcd-83348c633c5f"  
}
```